

**DESIGN AND IMPLEMENTATION OF AN IoT-BASED AIR
QUALITY MONITORING AND POLLUTION DETECTION
SYSTEM**

BY

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**A PROJECT REPORT SUBMITTED TO THE
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CERTIFICATION

This is to certify that the project upon which this report is based was carried out by **BAMGBADE ABDULLATEEF ADEDOYIN**, with matriculation number **199907** in the Department of Electrical & Electronic Engineering, Faculty of Technology, University of Ibadan.

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DEDICATION

I dedicate this report to Allah, my parents and the Department of Electrical and Electronic Engineering of the University of Ibadan for making this project a success.

ACKNOWLEDGEMENT

I would like to give thanks to Almighty Allah who has guided me all my life and given me the strength, endurance, and perseverance to complete this degree.

I would also like to appreciate Dr O.O Olakanmi for his support and guidance towards the successful completion of this project.

I would like to offer gratitude to my parents in the persons of Mr and Mrs. Bamgbade who have provided continued support and love all the years. I would like to appreciate my siblings: Adeolu, Adeola, and Olamide, who have always supported me in all endeavours.

Lastly, I would offer my appreciation to my friends who have supported me all the way.

ABSTRACT

Air pollution is a growing concern for public health and the environment, the existing air pollution levels in urban areas pose significant health risks to individuals. Major air pollutants include particulate matter, carbon monoxide, ozone, nitrogen, and sulphur dioxide. Air Pollution can affect both the environment and living things including plants and animals. In humans, air pollution can lead to respiratory afflictions like asthma and bronchitis or different forms of cancer in case of long-term exposure to pollutants. To solve these, an effective air quality monitoring and pollution detection system is needed to obtain the quantity of air pollutants within the immediate environment.

The system consists of an Arduino Nano board as an analog-to-digital converter (ADC), an ESP32 microcontroller, an I2C LCD display, an active buzzer, and multiple gas sensors, including the MQ7 sensor for CO, PMS5003 for PM2.5 and PM10, and Mics-6814 sensor for NO₂. The system collects air quality data from the sensors and sends it to the ThingSpeak.com web server for further analysis and visualization. The Arduino Nano board is used as an ADC to convert the analog signals from the gas sensors into digital signals, which can be processed by the ESP32 microcontroller. The ESP32 is used to read the sensor data and control the I2C LCD display and the active buzzer. The I2C LCD display shows the current air quality readings, and the active buzzer sounds an alarm if the air quality exceeds the predefined limits.

The system underwent testing in varying air quality conditions and demonstrates its ability to track variations in concentration of air pollutants effectively.

In conclusion, the developed air quality monitoring and pollution detection system is an effective solution to address the air pollution problem. It provides real-time monitoring of air quality and alerts the user if the air quality exceeds the predefined limits. The system can be easily scaled up or down depending on the requirements and can be used in various settings, such as homes, offices, and public places, to ensure a safe and healthy environment.

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CHAPTER 1

INTRODUCTION

1.1 Background to Study

Air pollution is the contamination of the indoor or outdoor environment by any chemical, physical, or biological agent that modifies the natural characteristics of the atmosphere. Household combustion appliances, motor vehicles, industrial facilities and fires originating from bush burning or garbage disposal are common sources of air pollution. Major air pollutants include particulate matter, carbon monoxide, ozone, nitrogen, and sulphur dioxide. Air Pollution can affect both the environment and living things including plants and animals. In humans, air pollution can lead to respiratory afflictions like asthma and bronchitis or different forms of cancer in case of long-term exposure to pollutants.

WHO data show that almost all the global population (99%) breathe air that exceeds WHO guideline limits and contains high levels of pollutants, with low- and middle-income countries suffering from the highest exposures. Household air pollution is generally a result of the use of inefficient stoves fuelled by kerosene or solid fuels like wood, charcoal, or coal. It could also be a result of using a fuel-powered generator in or around the household. Poor ventilation of homes can also lead to an increase in indoor air pollution. The combined effects of ambient(outdoor) air pollution and household air pollution are associated with 6.7 million premature deaths annually.

1.2 Problem Statement

Air pollution is a significant problem for Nigeria, with the country having a mortality rate of 307.4 for every 100,000 people as a result of air pollution according to the WHO. Long-term exposure to air pollutants can lead to stroke, heart disease, lung cancer, both chronic and acute respiratory diseases and death, while short-term exposure to air pollutants can trigger allergic reactions, cause irritation and discomfort and in the worst-case scenario death in air pollutants such as carbon monoxide (CO). These pollutants are generally unobservable to the senses which creates the need for a means of assessing the quality of air around us, enabling individuals to take steps to protect against exposure to the air pollutants.

1.3 Aim and Objectives

The aim of this project is to design and develop a reliable air quality monitoring system that provides remotely accessible real-time data on key air pollutants, such as particulate matter (PM_{2.5} and PM₁₀), nitrogen dioxide (NO₂), and carbon monoxide (CO), and general environment information like temperature and humidity, and alerts to harmful levels of pollution. To achieve this the following objectives must be carried out.

1. To develop a system that measures air pollutants and environmental parameters in real-time.
2. To develop a system that provides remote access to pollution data.
3. To develop an alarm system that can alert individuals to high levels of air pollution through an audible alarm.
4. To test and evaluate the performance of the system.

1.4 Scope of Project

The scope of this project is to design and implement a battery-powered air quality monitoring system that measures and displays real-time air quality data, including temperature, humidity, and concentrations of particulate matter, carbon monoxide (CO), nitrogen dioxide (NO₂) with the use of sensors and provides remote access to data through a web server and mobile application, as well as alert when the levels of the pollutants exceed recommended limits.

1.5 Organization of Report

The report is divided into five chapters. Each chapter is further divided into sections. Chapter one introduces the project, it gives the aim and objectives; and describes the problem to be solved. Chapter two is a literature review of the existing works it contains relevant information from credible sources presented to give a foundation for this project. Chapter three discusses the methodology used in the implementation of the project. Chapter four presents the results and observations during the testing of the deployed system. Chapter five gives recommendations on the project and future improvements.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview

Air quality monitoring refers to the process of measuring and assessing the levels of various pollutants in the air, including particulate matter, ozone, nitrogen dioxide, sulphur dioxide, and volatile organic compounds, among others. The goal of air quality monitoring is to identify the sources and levels of air pollution and to assess the potential health and environmental impacts of these pollutants. Air quality is a critical component of public health and environmental sustainability. With increasing urbanization and industrialization, air pollution has become a major global challenge, leading to numerous respiratory and other health problems. Therefore, air quality monitoring and alarm systems have become essential tools in managing air pollution levels and ensuring public health and safety. Air quality monitoring can be conducted using monitoring stations equipped with sensors and instruments that measure different parameters of air quality or by using machine learning based Air Quality Index (AQI) prediction models.

This literature review provides a comprehensive overview of the existing research and literature on air quality monitoring and alarm systems. The review aims to identify the current state of knowledge on the topic, including the major pollutants affecting the air quality, and the sensors used to detect the concentration of the pollutants, highlight any gaps in the literature, and suggest areas for future research. The review then explores the different types of air quality monitoring and alarm systems that have been previously implemented, designed, or proposed, including their features, benefits, and limitations. It discusses the various pollutants that can be detected by these systems.

2.2 Air Pollutants

Air pollutants are the substances in the air that can be harmful to human health, plants, and animals, and can also contribute to environmental problems such as climate change and ecosystem disruption. They can be both natural and human-made, originating from a variety

of sources, including household appliances, transportation, industry, agriculture, and natural sources like wildfires, dust storms, and harmattan.

There exist various air pollutants differing in chemical composition, reactive properties, health, and environmental effects, which can be grouped into four categories based on their similarities.

1. Gaseous pollutants (e.g., SO₂, NO_x, CO, O₃, Volatile Organic Compounds)
2. Persistent organic pollutants (e.g., dioxins).
3. Heavy metals (e.g., lead, mercury)
4. Particulate Matter.

According to WHO the pollutants of major public health concern include particulate matter, carbon monoxide, ozone, nitrogen dioxide and sulphur dioxide.

2.2.1 Particulate Matter

Particulate matter (PM) refers to the tiny particles that can be found in the air, these particles can be composed of a wide variety of materials, including dust, dirt, soot, and smoke, as well as pollutants like sulphate, nitrate, and organic compounds. Particulate matter can be produced by both natural and man-made sources, based on their origins they can be categorized into primary or secondary pollutants. The particulates emitted directly from their sources (construction sites, unpaved roads, fields, smokestacks, or fires) are known as primary particulates like sulphur dioxides and nitrogen oxides that are emitted from power plants, industries, and automobiles, while secondary particulates are formed by chemical reactions in the atmosphere.

The particles range between those less than 2.5 micrometres (microns) in diameter, referred to as PM_{2.5} and those between 2.5 and 10 micrometres in diameter, referred to as PM₁₀. Figure 2.1 shows a size comparison between PM_{2.5} and PM₁₀ particles, a grain of sand and a strand of human hair.

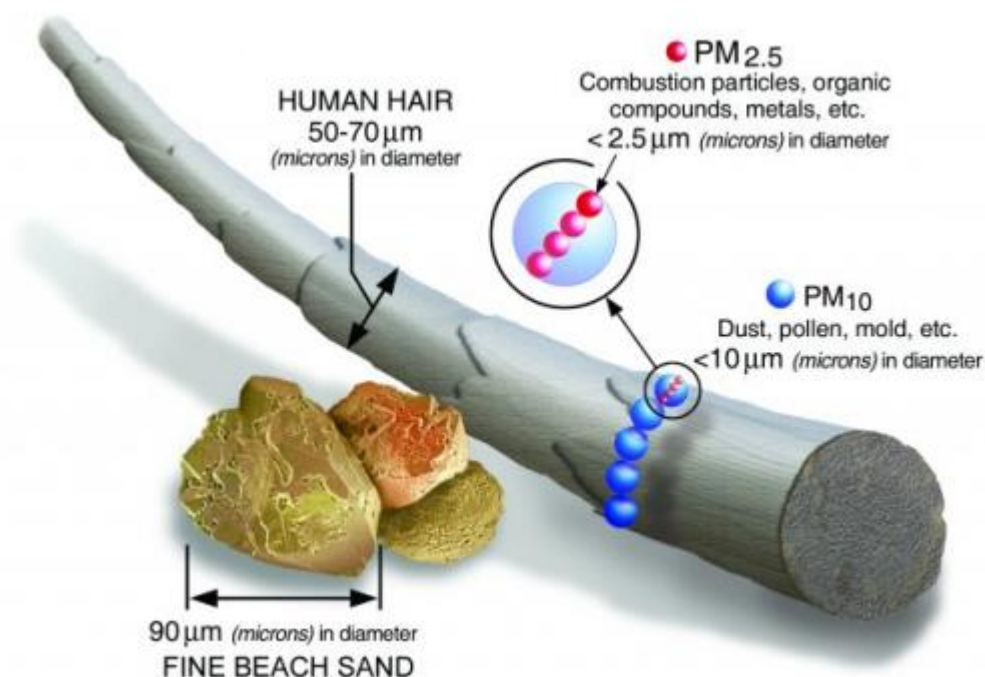


Figure 2.1: Size comparisons for PM particles

Particulate matter can be inhaled deeply into the lungs and can cause a range of health problems, including asthma, bronchitis, and heart disease. The health effects of particulate matter depend on the size and composition of the particles.

WHO updated global air quality guidelines for 2021 state that annual average concentrations of PM_{2.5} should not exceed $5 \mu\text{g}/\text{m}^3$, while 24-hour average exposures should not exceed $15 \mu\text{g}/\text{m}^3$, and PM₁₀ concentrations of $15 \mu\text{g}/\text{m}^3$ annual mean, $45 \mu\text{g}/\text{m}^3$ 24-hour mean.

2.2.2 Carbon Monoxide

Carbon monoxide (CO) is a colourless, odourless, flammable, and tasteless gas, produced by the incomplete combustion of organic materials, such as gasoline, natural gas, oil, coal and wood. Carbon monoxide is a highly toxic gas that may be harmful in low concentrations and deadly in high concentrations. When inhaled, it binds to haemoglobin in the blood, reducing the amount of oxygen that the body can absorb. This can lead to symptoms such as headaches, nausea, dizziness, confusion, and shortness of breath. At higher concentrations, it can cause loss of consciousness, coma, and even death. Sources of carbon monoxide

include gas stoves and ovens, furnaces and boilers, water heaters, fireplaces and wood stoves, and car exhaust.

WHO updated global air quality guidelines for 2021 recommend a 24-hour average concentration of Carbon monoxide of no more than 7 $\mu\text{g}/\text{m}^3$.

2.2.3 Ozone

Ground-level ozone, also known as tropospheric ozone (O_3), is formed when pollutants such as nitrogen oxides (NO_x) and volatile organic compounds (VOCs) react in the presence of sunlight. These pollutants are released by a variety of sources, including cars and trucks, industrial facilities, power plants, and consumer products such as solvents and paints. Exposure to ground-level ozone can have harmful effects on human health, particularly for people with respiratory problems such as asthma. It can cause shortness of breath, coughing, and chest pain, and can aggravate existing respiratory conditions.

WHO updated global air quality guidelines for 2021 recommend an 8-hour average concentration of Ozone of no more than 100 $\mu\text{g}/\text{m}^3$.

2.2.4 Nitrogen Dioxide

Nitrogen dioxide (NO_2) is a reddish-brown gas with an unpleasant and annoying odour. It primarily gets into the air from the burning of fuel. Nitrogen dioxide forms from emissions from cars, trucks and buses, power plants, and off-road equipment. Exposure to nitrogen dioxide can have harmful effects on human health. It can irritate the respiratory system, causing coughing, wheezing, and shortness of breath. It can also aggravate existing respiratory conditions such as asthma and increase the risk of respiratory infections. Long-term exposure to high levels of nitrogen dioxide can increase the risk of respiratory diseases and other health problems. Nitrogen dioxide also contributes to the formation of other air pollutants such as ground-level ozone and particulate matter.

WHO updated global air quality guidelines for 2021 state that annual average concentrations of Nitrogen dioxide should not exceed 10 $\mu\text{g}/\text{m}^3$, while 24-hour average exposures should not exceed 25 $\mu\text{g}/\text{m}^3$.

2.2.5 Sulphur Dioxide

Sulphur dioxide (SO₂) is a colourless gas with a pungent, irritating odour. SO₂ is the component of greatest concern and is used as the indicator for the larger group of gaseous sulphur oxides (SO_x). Sulphur dioxide is produced by the burning of coal, vehicle emissions and emissions from oil/gas fields and refineries. Short-term exposures to sulphur dioxide can lead to difficulty in breathing. People with asthma, particularly children, are sensitive to the effects of sulphur dioxide.

WHO updated global air quality guidelines for 2021 recommend a 24-hour average concentration of Sulphur dioxide of no more than 40 µg/m³.

2.3 Air Pollution Sensors

Over the past few years, there has been a rise in the accessibility and usage of affordable, user-friendly, and portable air pollution sensors, replacing the costly and intricate stationary equipment commonly employed in previous air quality monitoring practices.

Commercially available gas sensors are based on two main principles:

1. Chemical gas sensors utilize the interaction between gas components (e.g., nitrogen dioxide (NO₂), ozone (O₃), carbon monoxide (CO), volatile organic compounds (VOCs)) and sensing materials (Electrochemical Cell (EC) or Metal Oxide Semiconductor (MOS)) to detect and measure air pollutants.
2. Light-particle interaction is utilized in air quality monitoring, with visible and infrared light absorption measuring ozone (O₃), carbon dioxide (CO₂), and chemiluminescence detecting nitrogen dioxide (NO₂). Additionally, light scattering is employed to monitor particulate matter levels.

Figure 2.2 shows the general characteristics, including the strengths and considerations for the different types of pollutant sensors.

Gases	Strengths	Concern 1	Concern 2	Concern 3
Electrochemical cell (EC)				
CO, O ₃ , NO _x	Low cost, low power, small, real-time; more sensitive than MOS (metal-oxide-semiconductor) sensors	Interferences: CO, VOC, NO ₂	Drift, frequent recalibration needed, 1 yr lifetime	
Metal oxide semiconductor (MOS)				
CO, O ₃ , NO _x	Small size; stable, 1-2 yr lifetime, inexpensive	Sensitive to change of RH, T, P; cross-sensitivity	Power consumption; fragile materials	Typically less sensitive than EC
Non-dispersive infrared absorption (4.26 μ m)				
CO ₂	Compact, stable to changing RH and T	Sensitivity depends on path length	Calibration may be misinterpreted or inaccurate	Some single beam devices auto-calibrate as if background CO ₂ is 400 ppb
Ultraviolet absorption (254 nm)				
O ₃	Accuracy, stable to change in P	Size (not yet miniaturized)	Sensitive to changes in relative humidity	Cost
Particle Properties ^a				
Light Scattering	Small, inexpensive, commercially available	Not a direct mass measurement	Does not measure ultrafine particles	
Light Absorption	Handheld well established, stable, continuous	Still relatively large and costly	Requires changing a filter	
Direct Particle Mass	Small, inexpensive, direct mass concentration; FBAR, QCM	In development stage	Likely sensitive to changes in T and RH	

Figure 2.2: Characteristics of Gas and PM Sensors

MOS sensors are widely utilized for measuring major gaseous pollutants such as carbon monoxide, ozone, nitrogen dioxide, and sulphur dioxide. These sensors are favored for their compact size, low power consumption, affordability, fast response, and high sensitivity. The latest MOS gas sensors are integrated into small-sized packages with digital communication interfaces for improved functionality.

To measure particulate matter (PM), optical particle counters that rely on light scattering are employed. These counters utilize a light-emitting diode as the light source and incorporate a small fan within the module to draw particles into the sensing area.

2.4 The Air Quality Index

The Air Quality Index (AQI) is a metric utilized for communicating air quality levels and associated health risks caused by pollution. It is commonly employed by government agencies and organizations to inform the public about present and predicted air quality conditions, release air quality alerts, and provide guidance on mitigating exposure to air pollution.

The United States Environmental Protection Agency (US EPA) defines the national outdoor air quality standards based on the six common air pollutants, namely particulate matter (PM₁₀ and PM_{2.5}), carbon monoxide (CO), ozone (O₃), sulphur dioxide (SO₂) and nitrogen dioxide (NO₂) and the index was calculated using the following formula.

Equation 2.1

$$I_p = (C_p - BP_{low}) \times \frac{I_{high} - I_{low}}{BP_{high} - BP_{low}} + I_{low}$$

Where:

I_p = Index value for pollutant p

C_p = Rounded concentration of pollutant p

BP_{high} = Higher Breakpoint value of C_p

BP_{low} = Lower Breakpoint value of C_p

I_{high} = Index Breakpoint value of BP_{high}

I_{low} = Index Breakpoint value of BP_{low}

The breakpoint and index values used in the equation 2.1 are gotten from the air pollutant concentration data and breakpoint table shown in Table 2.2 below.

Table 2.1: US EPA's Breakpoint and AQI Index

O₃ (ppm)	PM10 (µ/m3)	PM2.5 (µ/m3)	CO (ppm)	SO₂ (ppm)	NO₂ (ppm)	AQI Values	Level of Health Concern
0.000 - 0.059	0 - 54	0.0 - 15.4	0.0 - 4.4	0.000 - 0.034	-	0 - 50	Good
0.060 - 0.075	55 - 154	15.5 - 40.4	4.5 - 9.4	0.035 - 0.144	-	51 - 100	Moderate
0.076 - 0.095	155 - 254	40.5 - 65.4	9.5 - 12.4	0.145 - 0.224	-	101 - 150	Unhealthy for Sensitive Groups
0.096 - 0.115	255 - 354	65.5 - 150.4	12.5 - 15.4	0.225 - 0.304	-	151 - 200	Unhealthy
0.116 - 0.374	355 - 424	150.5 - 250.4	15.5 - 30.4	0.305 - 0.604	0.65 - 1.24	201 - 300	Very Unhealthy
-	425 - 504	250.5 - 350.4	30.5 - 40.4	0.605 - 0.804	1.25 - 1.64	301 - 400	Hazardous
-	505 - 604	350.5 - 500.4	40.5 - 50.4	0.805 - 1.004	1.65 - 2.04	401 - 500	Hazardous

The index is presented on a scale ranging from 0 to 500 and is divided into six categories, each assigned specific color-coded and numerical ranges: Good (0-50), Normal (51-100), Unhealthy for sensitive groups (101-150), Unhealthy (151-200), Very unhealthy (201-300) and Hazardous (>300).

To calculate the AQI, each pollutant is measured either at a 1,8 or 24-hour average depending on the type of pollutant. Then, the average value from each pollutant was converted into an individual index based on the breakpoint table. If multiple pollutants are measured at a monitoring site, then the largest AQI value is reported for the location.

2.5 Review of Related Works

There is a growing demand for effective air quality monitoring stations to provide real-time data on air pollution levels and enable prompt response to potential risks, over the years researchers have taken multiple approaches to air quality monitoring and pollution sensing, the two major ways are:

1. Sensor-based air quality monitoring: this involves the use of specialized sensors interfaced with microcontrollers to detect and measure levels of air pollution in real-time.
2. Machine learning-based air quality monitoring: this involves the use of algorithms and models to analyse data from air quality sensors and other sources, in order to predict and monitor air pollution levels. Machine learning techniques can be used to process large amounts of data from multiple sources, such as satellite imagery, weather data, and traffic data, to provide real-time updates on air quality.

2.5.1 Sensor-Based Air Quality Monitoring

Sensor-based air quality monitoring systems use various types of sensors to detect and measure air pollutants, these sensors are often combined with microcontrollers, wireless communication technologies, and data analysis tools to create a comprehensive air quality monitoring system. Researchers have worked on various sensor-based air quality monitoring systems making use of different microcontrollers, and sensors.

The Arduino microcontroller has been used by a lot of researchers in recent times. Al Ahasan et al. (2018) describes an Arduino-based real-time air quality and pollution monitoring system. The hardware comprised an Arduino UNO microcontroller, an MQ-135 air quality sensor capable of detecting carbon dioxide, Benzene, Alcohol, smoke, nitrogen dioxide and ammonia, and an LCD display to display the collected data. The system uses self-developed customised software running on the Arduino Integrated Development Environment (IDE) for real-time data gathering from the hardware system. The system is tested within different air-polluted conditions, using cigarette smoke, mosquito coil burning smoke and motor vehicle smoke, providing the concentration of pollutants in ppm (parts per million) within a 3-meter range at 2-second intervals. The system has the following limitations, it is unable to measure particulate matter, ozone, carbon monoxide, and sulphur dioxide, which are part of the WHO major pollutants, additionally, the collated pollution data is only available on the LCD display of the system, and finally, the alarm system is limited to a visual message on the LCD display of the system.

Similarly, Kaur et al. (2016) developed an air quality monitor based on the Arduino microcontroller that monitors environmental parameters such as temperature and humidity

and concentration of pollution including carbon monoxide, carbon dioxide, smoke, alcohol, and LPG. The hardware consists of an Arduino Uno microcontroller, and gas sensors: MQ-135 for carbon dioxide (CO₂), MQ-2 for Liquefied Petroleum Gas (LPG), alcohol and smoke, MQ-7 for Carbon monoxide (CO), a Zigbee S2 module, a GSM module, and a Bluetooth module. The software consists of C language code running on the Arduino IDE and X-CTU simulation environment serving as the base station receiving information from the Zigbee module. The data in the air is acquired by sensors and processed by the microcontroller, the Zigbee S2 acts as a gateway for the communication between the microcontroller and the base station. The GSM module is used to send a text message whenever the value of carbon monoxide exceeds a safe limit. Temperature and humidity values are also transmitted via short-range communication with the Bluetooth module.

Shah et al. (2018) proposed the design of an IOT-based air pollution monitoring system which monitors the air quality over a web server using the internet and will trigger an alarm when the air quality goes beyond a certain level. The system will monitor the following pollutants: carbon dioxide, smoke, alcohol, benzene, ammonia, nitrogen oxides and Liquefied Petroleum Gases (LPG), and temperature and humidity. The hardware consists of an Arduino Uno microcontroller, gas sensors MQ6 for LPG and MQ135 for the other pollutants to obtain the concentration of the pollutants in the air, an LM35 temperature sensor, SY-HS-220 humidity sensor, an ESP8266 Wi-Fi module to give the system access to the internet, a GSM module to enable communication with a GSM system and a buzzer to signal when the concentration of pollutants goes beyond a certain level. The software would consist of code written in C running on the Arduino IDE. The system would display the concentration of the pollutants in parts per million (ppm), along with a message on the LCD and web page indicating if the air is fresh or not along, and in the case of poor air quality the buzzer begins beeping and an alert message is sent to a phone through GSM.

Jha (2020) presented an advanced real-time air quality reporting system supported by Internet of Things (IoT) architecture, that measures the concentration of PM_{2.5}, carbon monoxide and nitrogen dioxide in the air and integrates the real-time data into a custom-designed mobile application. The hardware consists of an Arduino Uno microcontroller, gas sensors: MQ135, MQ7 and dust sensor GP2Y1010AU0F, a Wi-Fi module ESP8266 and a buzzer. The software architecture consists of code written in C running on the Arduino IDE

for real-time processing and gathering of data from the sensors and a mobile phone application developed on Android Studio. The sensors measure the concentration of the pollutants which is then converted to ppm and then using the ppm of the pollutants the Air Quality Index (AQI) is calculated, the AQI data is then sent to the thingspeak.com web server and with the use of API keys the data is sent to the Android Mobile Application which categorizes the air based on the air quality data and the AQI of that area into Good, Bad and Severe, and displays it along with the ppm of the pollutants and the AQI.

Alternatively, due to its versatility, the Raspberry Pi is also used by researchers. Kumar and Jasuja (2017) presented a real-time standalone air quality monitoring system using ARM-based microcomputer Raspberry Pi and a sensor to measure parameters including PM 2.5, carbon monoxide, carbon dioxide, temperature, humidity, and air pressure, making use of the Internet of Things (IoT) and cloud computing for better management of collated data. The hardware consists of a raspberry pi as the major control node, the pollutants sensors including DSM501A used for measuring particulate matter, MHT22 for temperature and humidity, BMP180 for pressure, MQ9 and MQ135 are connected to an Arduino Uno board which is interfaced with the Raspberry pi through a USB cable. A Wi-Fi adapter is used for providing internet access to the system. Software architecture consists of Node-RED for IOT integration, Arduino IDE to program the Arduino board and the protocol MQTT (Message Queuing Telemetry Transport) for enabling communication between the sensors and clients. IBM cloud services are used for the storage and management of data. The device is tested, with measured values very close to expected values indicating the accuracy of the system with the data being available on the IBM Watson platform allowing for access through the internet with the use of an IBM Bluemix account. The device however makes no provision for an alarm to alert to high pollution levels.

Similarly, Kiruthika and Umamakeswam (2017) designed a low-cost pollution control and air quality monitoring system using Raspberry Pi for Internet of Things that uses sensors to collect data on the surrounding environment, and a prediction algorithm based on the collected data is used to proactively implement a solution to any problem encountered. The hardware consists of a Raspberry Pi single board computer, used as a sensor node and gateway node, an ESP8266 Wi-fi module for adding Wi-Fi functionality to the Pi, an MQ5 gas sensor, a temperature and humidity sensor, a soil moisture sensor and a GrovePi device

to interface the sensors with the main board controller. The software comprises of python code used for the Raspberry Pi, a MySQL database for storing collected data that can be queried whenever required, and ThinkSpeak open data IoT for evaluation of results with data analytics. The system uses four sensors to measure the humidity, temperature, gas density and soil moisture, a threshold for each parameter, the measured values are continually recorded and at a specified interval (three, five or ten minutes) are transmitted to an online database. The system sends an alert to the user through a connected website when any of the sensed parameters exceed their corresponding threshold values. The system is proposed to be installed in several places of preferably higher population density to observe the parameters over a period, this data will be highly useful for understanding the levels of pollution in a place.

Some researchers though have chosen to make use of other microcontrollers. Marinov et al. (2016) presented Outsense, an approach for cost-effective measurement of relevant environmental parameters, based on a scalable sensor array with integrated amperometric and infrared gas sensors. The hardware consists of microcontroller PIC18F87K22, a GPS module based on the MTK3339 chipset, ESP8266 chip as a Wi-Fi module, gas sensors: CO-AF (for carbon monoxide), NO2-A42F (for nitrogen dioxide), OX-A421 (for ozone) and O2-A2 (for oxygen), and sensors for physical parameters: MPXA6115A (for pressure), HDC1050 (for humidity), and MCP9808 (for temperature). The software architecture consists of firmware written using C programming language and a web server.

Gunawan et al (2018) designed and implemented a portable outdoor air quality measurement system using Arduino capable of measuring the concentration of carbon monoxide, ground-level ozone, and particulate matter (PM10 and PM2.5) in the air and converting the readings to Air Pollutant Index (API) value. The hardware consists of an Arduino Uno microcontroller, MQ-9 CO sensor, MQ-131 O3 sensor, GP2Y1010AUF PM10 sensor, Shinyei PPD42NS PM2.5 sensor, 16x2 LCD display and an SD card reader for data logging. The device is powered by a 5V external power supply. The software is programmed using the Arduino Integrated Environment (IDE) and is divided into four major parts data collection, data conversion, data comparison and data classification. Each sensor is subjected to experiments for evaluation. The CO sensor is tested under two conditions, smoke from a cigarette and emission from car exhaust. The PM10 and PM2.5 are tested using powder as a

dust source and a fan used to spread the powder in a container. The ozone sensor is tested near a bus stop beside a road with light traffic at two different times, in the morning and the afternoon. The overall system is tested at three different locations to measure the air quality and the results are compared to the API values measured by Continuous Air Quality Monitoring stations (CAQMs) in Batu Muda, Malaysia at the same time of the experiment. The comparison reveals a percentage error of 14%. The system is shown to be a portable, cost-effective efficient air quality monitor that can be used anytime and anywhere, it however lacks a pollution detection or alarm system. using a Linux distribution operating system and MySQL database.

Abiduzzaman et al. (2021) developed a real-time outdoor air quality monitoring system based on the ESP32 microcontroller that measures the concentration of carbon monoxide (CO) and nitrogen dioxide (NO₂) in the air and enables access to the Air Pollution Index (API) and location data through a mobile application. The hardware consists of the gas sensors: MQ-7 for CO and MQ-135 for NO₂ and a GPS module which are connected to an ESP32 microprocessor. The software consists of a mobile application built using Flutter and Firebase as the storage database. The ESP32 processes the pollution data received from the sensors and the location data received from the GPS module, the collated information is then sent to the Firebase database with the aid of the Wi-Fi connectivity feature of the microcontroller, the mobile application then accesses the information from the database through APIs. The system is tested at various locations and the API information is compared to the API of the locations available online from (AQICN, and IQAir) with an accuracy of 92.3% proving the efficiency of the system.

Phala et al. (2016) developed an air quality monitoring system based on IEEE/ISO/IEC 21451 standards which explores the implementation criteria of transducer independent interface (TIM) in the devices and proposed the format criteria of the transducer electronic data sheet (TEDS). The hardware of the system consists of electrochemical sensors: EC4-500-CO, EC4-20-SO₂, EC4-20-NO₂, infrared CO₂ sensor, a 16-bit dsPIC30F4013 microcontroller with modified Harvard architecture, and a GSM module. The software consists of a Graphical User Interface (GUI) developed using MATLAB. The microcontroller receives data from the sensors and then communicates serially with the GSM module using UART. The gas concentration data is sent to the GSM where it is transmitted

wirelessly. The GUI receives the data from the GSM module and the data is stored in a data text file.

2.5.1.1 Advantages of Sensor-Based Air Quality Monitoring

1. Sensor-based air quality monitors can provide real-time data on air pollution levels.
2. Sensor-based air quality monitors can be designed to be portable, allowing them to be deployed in a variety of locations for short-term or long-term monitoring.
3. Sensor-based air quality monitors are often much less expensive than traditional air quality monitoring systems, making them more accessible.

2.5.1.2 Limitations of Sensor-Based Air Quality Monitoring

1. The accuracy of sensors used can vary widely and may not be as reliable as professional-grade monitoring systems.
2. The sensors have a limited lifespan and may need to be replaced frequently.
3. Sensor-based air quality monitors have a limited range and may not be able to cover large areas.

2.5.2 Machine Learning-Based Air Quality Monitoring

Machine learning-based air quality monitoring is a process of analysing and predicting air quality levels using data collected from various air quality monitoring stations. This approach uses machine learning algorithms to process and analyse the data, identify patterns and trends, and predict future air quality levels. Simu et al. (2020) proposed a method of air pollution prediction that uses Machine Learning (ML) models with pollution data to predict future emission parameter values. The module makes use of python generated dummy pollution data, and machine learning algorithms including K-Nearest Neighbor (KNN), Support Vector Regression (SVR), Random Forest (RFS), Multi-Linear Regression (MLR) and Multi-Layer Perceptron (MLP) then implemented on the dummy data to predict the value of emission rate in m^3/hr and wind velocity. The accuracy of each case is then checked by measuring the mean square error, algorithms are then optimized to reduce the error thereby increasing the accuracy of the prediction. and the best algorithm, the one with the lowest error is selected which turned out to be the Multi-Layer Perceptron algorithm, and

finally, the extent of pollution spread was calculated using the Gaussian model to calculate dispersion.

2.5.2.1 Advantages of Machine Learning-Based Air Quality Monitoring

1. By analysing historical air quality data, machine learning algorithms can make predictions about future air quality conditions.
2. Machine learning algorithms can adapt to changing conditions, such as weather patterns, and adjust air quality predictions accordingly. This allows for more accurate and responsive air quality monitoring.
3. Machine learning algorithms can process large amounts of data quickly and efficiently and identify patterns and trends leading to more accurate and reliable air quality measurements.

2.5.2.2 Limitations of Machine Learning-Based Air Quality Monitoring

1. Machine learning algorithms require large amounts of high-quality data to generate accurate predictions. However, in some areas, data may not be available or may be of poor quality, which can limit the accuracy of machine learning predictions.
2. Machine learning-based air quality monitoring systems can be more expensive than traditional monitoring methods, as they require advanced computational resources and expertise to operate.
3. Machine learning algorithms can be complex and difficult to understand, which can make it challenging to interpret the results and communicate them effectively to regular users.

2.6 Conclusion

In conclusion, air pollution is a serious issue that affects the health of people and the environment. An air quality monitoring and alarm system can help mitigate the negative effects of air pollution by providing real-time information about air quality levels and alerting users when levels exceed safe limits. Through the literature review, we have explored the different components and technologies involved in air quality monitoring systems, as well as the benefits and challenges associated with their implementation. The

review has shown that there are various types of sensors available for measuring air quality parameters such as particulate matter, gases, and humidity. These sensors can be integrated into a monitoring system that collects, processes, and displays the data. Wireless communication and cloud-based platforms can provide easy access to data and enable remote monitoring and control.

Comparing the two methods of air quality monitoring, while machine learning-based air quality monitoring systems offer advanced data analysis and predictive capabilities, they may be more suitable for environments with ample computational resources and expertise. In Nigeria, the practicality, simplicity, real-time monitoring, power efficiency, and cost-effectiveness of microcontroller-based systems make them a more suitable choice for air quality monitoring applications.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter will explain the methodology of the design and implementation of the IoT-based air quality monitoring and pollution alarm system including the system architecture, hardware and software requirements, the schematic setup and the algorithm used in setting up the functionality of the system.

3.2 System Architecture

The block diagram in Figure 3.1 visually illustrates the interconnections and relationships among the different components comprising the system, effectively portraying the functional structure and dependencies of the system's architecture.

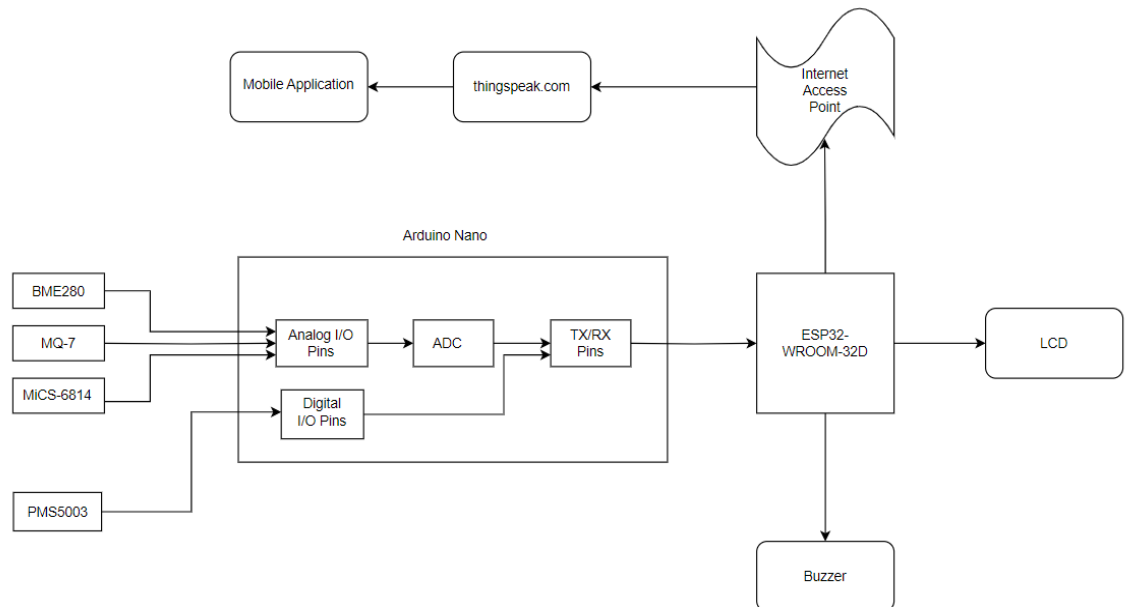


Figure 3.1: Block Diagram for Air Quality Monitoring System with Pollution Alarm

3.3 Hardware Unit

The hardware unit of the air quality monitoring system includes sensors for measuring pollutants, a microcontroller for data processing, and output devices such as displays. It enables real-time data collection, analysis, and visualization for effective air quality monitoring.

Figure 3.2 depicts the schematic setup of the air quality system, showcasing the interconnection of sensors, microcontrollers, and output devices for comprehensive monitoring and analysis.

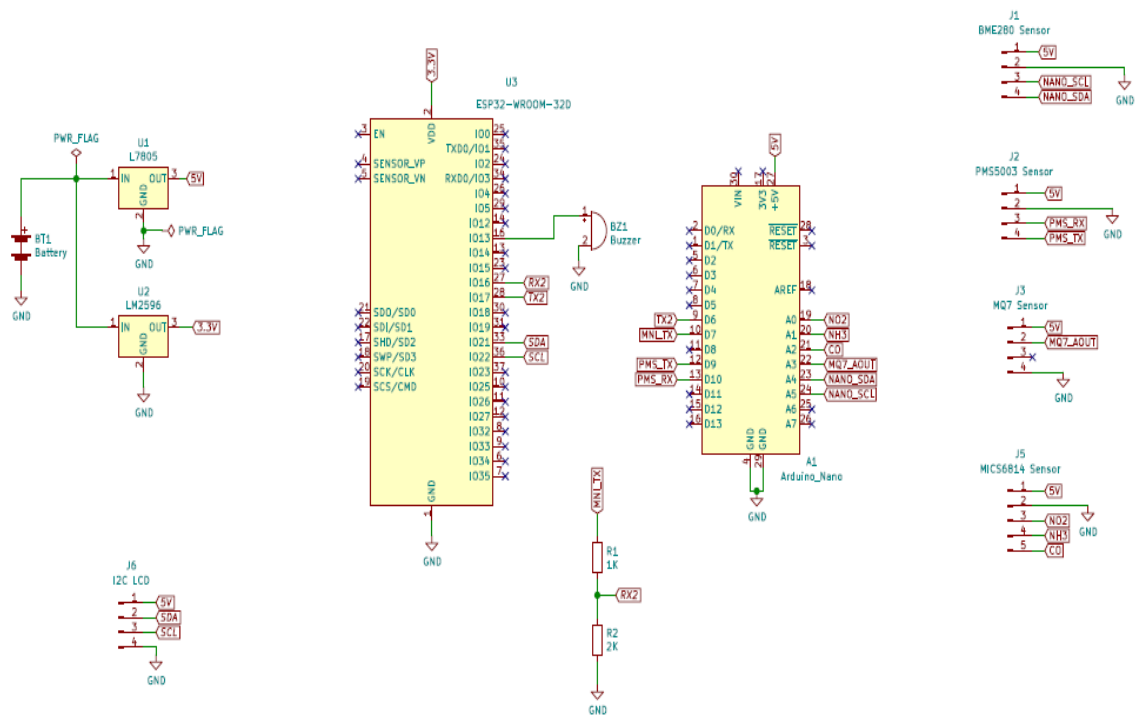


Figure 3.2: Schematic Setup for Air Quality Monitoring System with Pollution Alarm

The hardware unit of the air quality monitoring system is divided into 4 sections each encompassing a function. These sections are listed below.

1. Power Section
2. Sensor Section
3. Microcontroller Section
4. Output Section

3.3.1 Power Section

The power section is responsible for providing a stable and appropriate power supply to the components of the monitoring station. It consists of a pair of 3.7V rechargeable lithium-ion (Li-ion) batteries and a voltage regulator circuit. The Li-ion battery is used because of its high energy density, low self-discharge rate, and long cycle life.

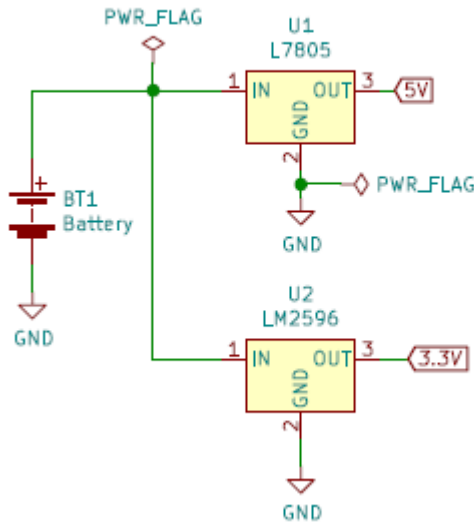


Figure 3.3: Schematic Circuit of Power Section

A voltage regulator circuit is used to regulate the Li-ion battery voltage to a stable 3.3V or 5V, which is required by the Arduino Nano and ESP32, respectively. For the Arduino Nano, The L7805 voltage regulator is used, the L7805 is a three-terminal linear voltage regulator that provides a fixed output voltage of 5V from an input voltage of up to 35V. The L7805 is connected to the input voltage source, ground, and output load, which is the Arduino Nano. For the ESP32 connection, the LM2596 voltage regulator is used, The LM2596 is a popular step-down voltage regulator that can be used to convert a higher input voltage to a lower output voltage with high efficiency. It accepts the input voltage and outputs a fixed voltage of 3.3V, the operating voltage of the ESP32. The LM2596 is connected to the input voltage source, ground, and output load.

3.3.2 Sensor Section

The sensor section of the hardware unit consists of the sensors used to measure the concentration of the pollutants and the Arduino Nano microcontroller, which acts as the Analog-to-Digital converter. Table 3.1 shows the sensors used and their measurement parameters.

Table 3.1: List of Sensors

Sensor	Parameter
BME280	Temperature and Humidity
PMS5003	PM2.5 and PM10
MQ-7	Carbon monoxide (CO)
MiCS-6814	Nitrogen dioxide (NO2)

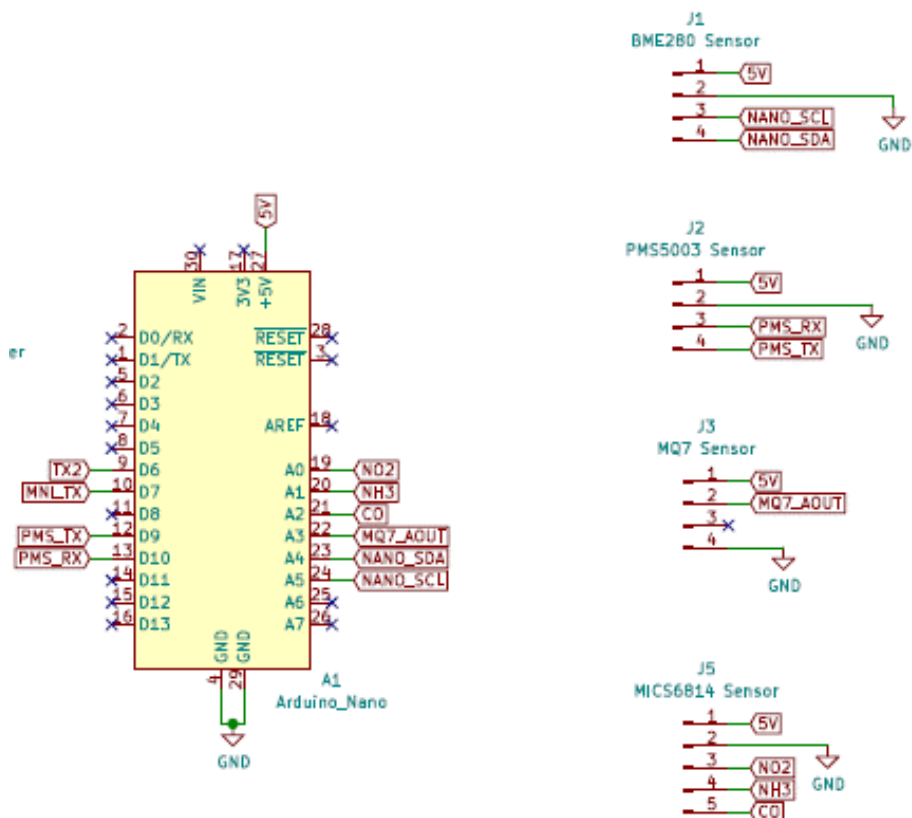


Figure 3.4: Schematic Circuit of Sensor Section

logic level while the Arduino Nano is at a 5V logic level. A voltage divider circuit is employed for the step-down operation.

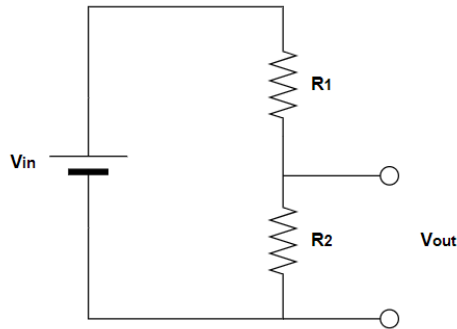


Figure 3.6: Voltage Divider Circuit Diagram

$$V_{out} = V_{in} \frac{R_2}{R_1 + R_2}$$

$$\frac{R_2}{R_1 + R_2} = \frac{V_{out}}{V_{in}}$$

$$V_{in} = 5V \text{ (logic level of Arduino Nano)}$$

$$V_{out} = 3.3V \text{ (logic level of ESP32)}$$

$$\frac{R_2}{R_1 + R_2} = \frac{3.3}{5}$$

$$5R_2 = 3.3(R_1 + R_2)$$

$$5R_2 - 3.3R_2 = 3.3R_1$$

$$1.7R_2 = 3.3R_1$$

$$\frac{R_2}{R_1} = \frac{3.3}{1.7} \approx 2$$

$$\therefore R_2 : R_1 = 2 : 1$$

Using the derived ratio requirement, $1k\Omega$ and $2k\Omega$ resistors are chosen for R_1 and R_2 respectively. The implemented voltage divider circuit is shown in Figure 3.7 below.

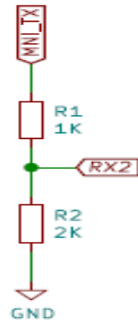


Figure 3.7: Schematic Circuit of Implemented Voltage Divider

3.3.4 Output Section

The output section of the system comprises devices that provide visual and auditory feedback to the user. These devices include an I2C LCD module, that uses the I2C communication protocol to interface with a microcontroller and is used for displaying sensor data, and an active buzzer will act as an alarm when the AQI threshold is exceeded.

As shown in Figure 3.8, the LCD is connected to the ESP32 microcontroller as follows:

1. GND to GND
2. VCC to 3.3V
3. Serial data (SDA) to GPIO21 pin
4. Serial clock (SCL) to GPIO22 pin

The buzzer is connected to the GPIO13 pin and to GND.

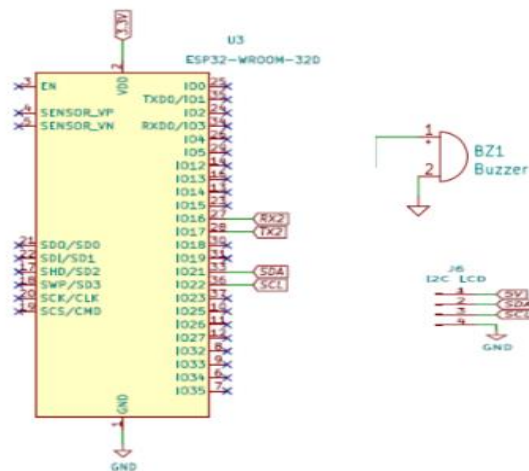


Figure 3.8: Schematic Circuit of Output Section

3.4 Software Unit

The software unit of the air quality monitoring station is an integral part of the overall system. It involves the implementation of code that runs on the microcontroller using the Arduino IDE (Integrated Development Environment). The Arduino IDE provides a user-friendly platform for writing, compiling, and uploading code to the microcontroller.

To transmit the collected data, two options are used: ThingSpeak and MQTT. ThingSpeak is a cloud-based platform that enables easy data collection, visualization, and analysis. The air quality data can be sent to the ThingSpeak server, which then allows you to monitor and analyze the data through its web interface or API. MQTT (Message Queuing Telemetry Transport) is a lightweight messaging protocol commonly used for IoT applications. It provides a publish-subscribe model, allowing devices to send and receive messages through an MQTT broker. In the context of an air quality monitoring station, the data collected by the sensors can be published to an HiveMQ broker, on a topic named Air-monitor, which can then distribute the data to various subscribers for further processing or visualization.

By leveraging the Arduino IDE, ThingSpeak, and MQTT, the software unit of the air quality monitoring station facilitates data collection, storage, and communication, enabling real-time monitoring, analysis, and integration with other systems or platforms.

3.4.1 Algorithm for Functioning of the System

The underlying operations of the system can be explained by referring to the flowchart illustrated in Figure 3.9. When the system is started, the sensors begin continuously monitoring the air quality and sending the data to the ADC for conversion to digital signals. The ESP32 microcontroller processes the digital signals and calculates the AQI value based on the concentration of pollutants detected using Equation 2.1 and Table 2.2, the actual AQI is set to the maximum calculated value. If the AQI value exceeds the threshold which is set to 200, the alarm is triggered to alert users to the unsafe air quality. The data is also displayed on the LCD and uploaded to Thingspeak for monitoring and analysis.

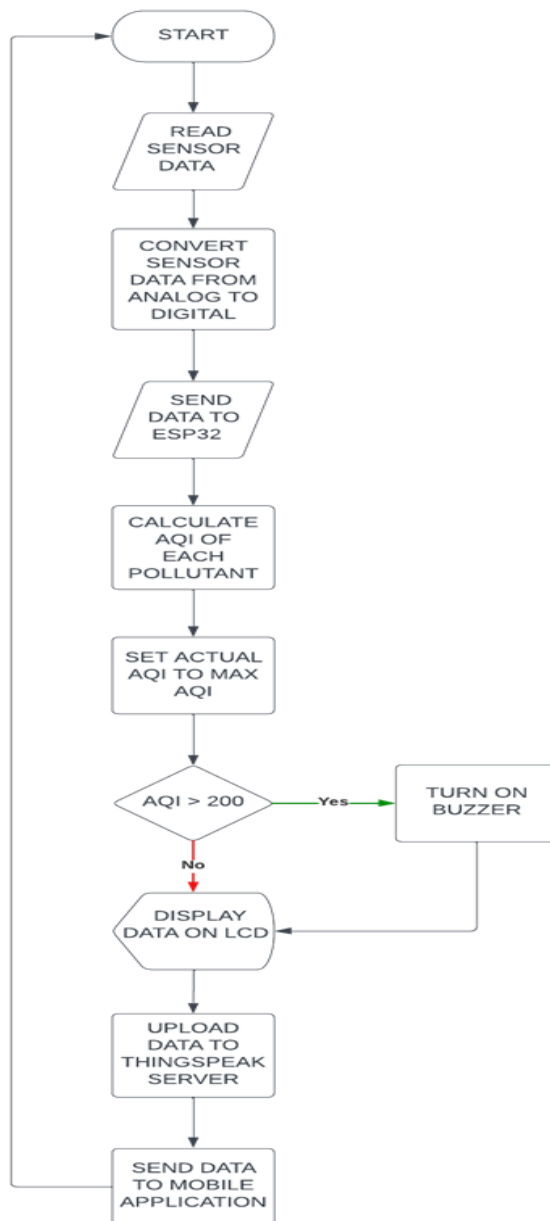


Figure 3.9: Flowchart of Working Principle of Air Quality Monitoring System

3.4.2 Algorithm of Remark Generation

The process of generating the air quality remark is determined utilizing the flowchart presented in Figure 3.10.

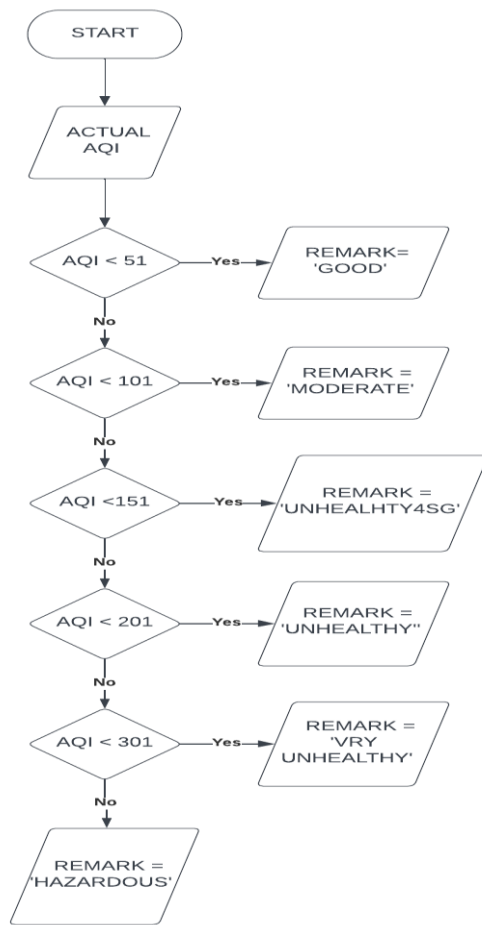


Figure 3.10: Flowchart of Remark Generation

The remark is selected using the level of concern column of Table 2.1. With “UNHEALTHY4SG” and “VRY UNHEALTHY” representing “UNHEALTHY FOR SENSITIVE GROUPS” AND “VERY UNHEALTHY” respectively due to limitation of space in the LCD.

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 Introduction

This chapter aims to interpret and discuss the data collected during the implementation and testing phases of the project.

The system is subjected to two forms of testing:

1. Operational Testing: In which the functionality of the system is evaluated against the objectives proposed towards achieving the aim of the project.
2. Performance Testing: In the ability of the system to capture changes in air quality parameters is evaluated.

4.2 Operational Testing

The system is subjected to evaluation to assess its capability in meeting the project objectives and to verify its overall functionality and performance alignment with the project's aim. Three key functionalities must be present in the system to indicate the successful accomplishment of the project objectives.

1. Real time measurement.
2. Remote access.
3. Alarm system.

4.2.1 Real Time Measurement

The system enables real-time monitoring of pollutant concentrations such as PM_{2.5}, PM₁₀, CO, and NO₂, as well as environmental parameters including temperature and humidity. This functionality allows users to conveniently access and view the collected data through the LCD display as shown in Figure 4.1

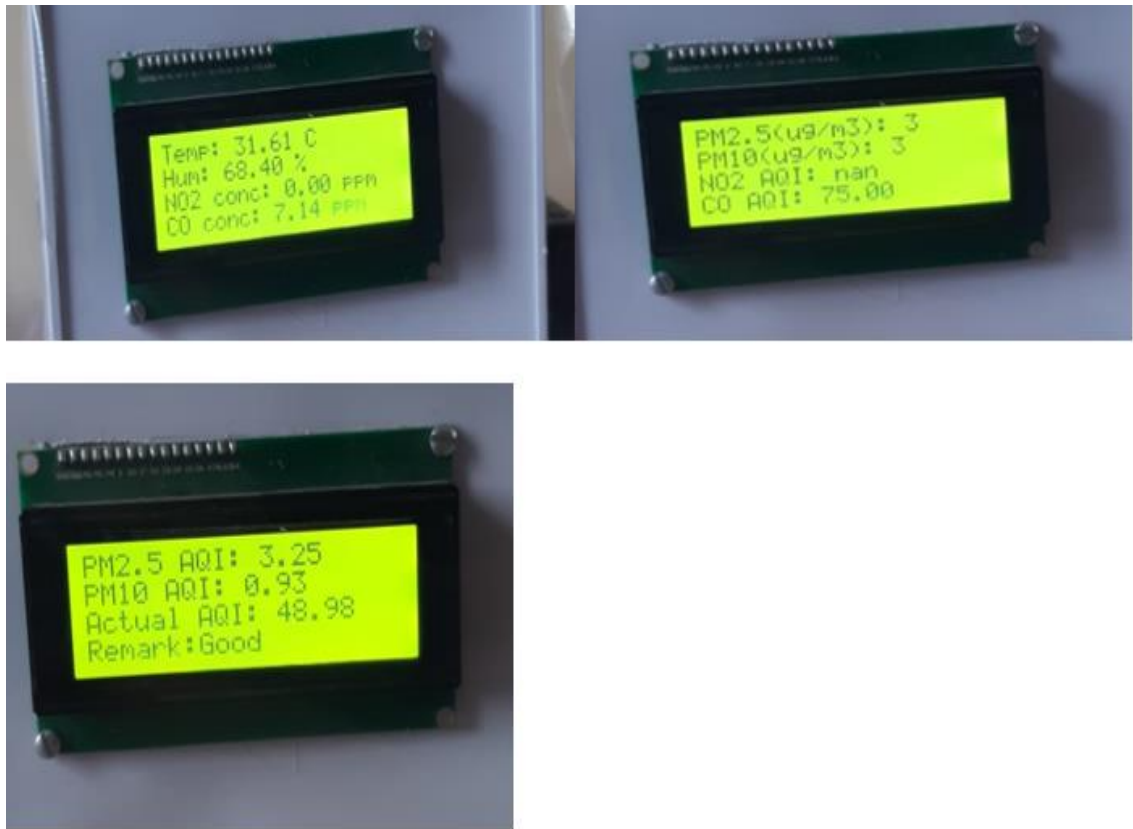


Figure 4.1: Display Showing Real Time Data

4.2.2 Remote Access

Users can effortlessly retrieve the system's data through various channels, such as the ThingSpeak website, application, and an MQTT dashboard. Accessing the data through the MQTT dashboard involves subscribing to the Air-monitor topic hosted on HiveMQ, as depicted in Figure 4.2. Alternatively, the data can be accessed through the ThingSpeak channel, as illustrated in Figure 4.3.

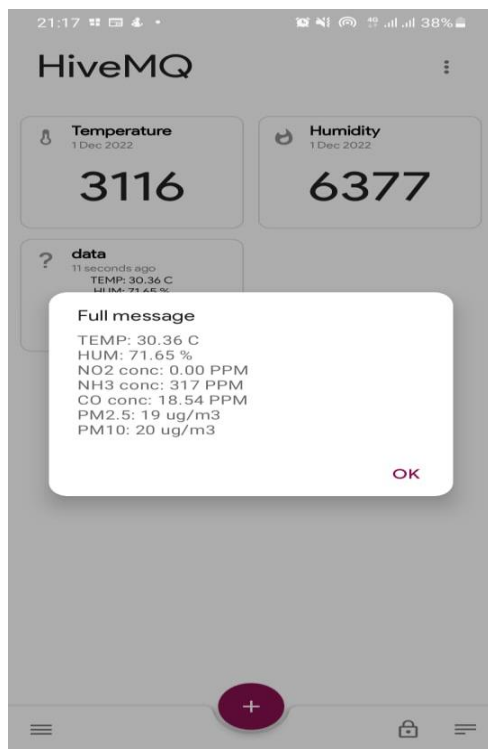


Figure 4.2: View of Data on Mobile Application

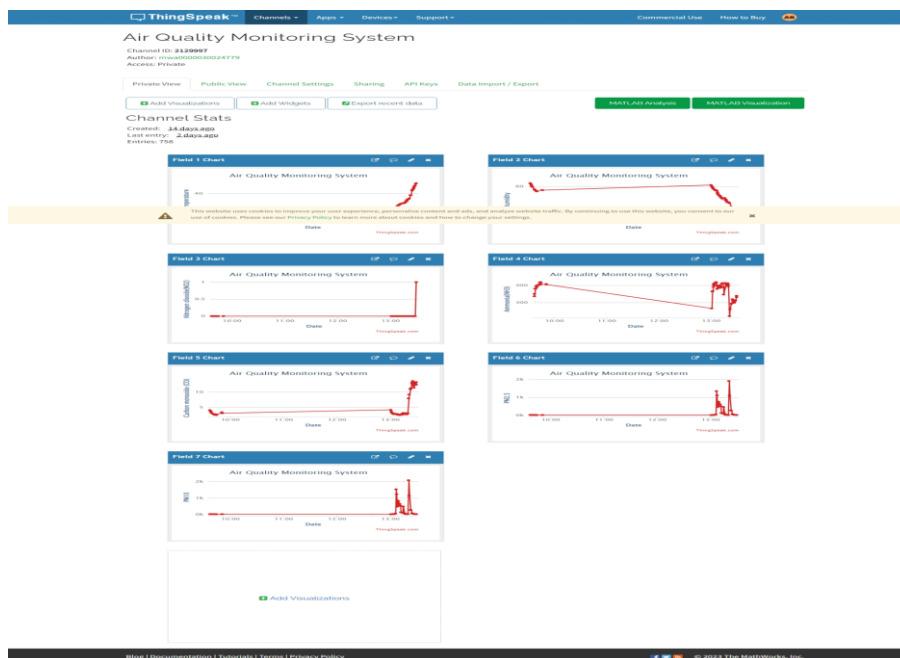


Figure 4.3: Overview of Thingspeak Channel

4.2.3 Alarm system

The buzzer is triggered when the actual Air Quality Index (AQI) exceeds the predefined threshold of 200, initiating its activation state. It remains active until the actual AQI value decreases to or falls below the specified threshold, providing an audible indication of poor air quality until it improves to a satisfactory level.

4.3 Performance Testing and Evaluation

The system undergoes testing in diverse air quality environments to assess its ability to accurately monitor fluctuations in air quality.

The Test environments are as follows:

1. Sealed enclosure environment
2. Humidity chamber environment
3. Smoke box environment
4. Carbon monoxide exposure environment

4.3.1 Sealed Enclosure Environment

The system is enclosed within a flexible rectangular storage bag to simulate a controlled environment with minimal levels of suspended particulate matter (PM) for a duration of 15 minutes, while measurements are recorded at 30-second intervals.

Based on the data presented in Figure 4.4, it is evident that the Air Quality Index (AQI) for PM_{2.5} and PM₁₀ exhibits a gradual decline, eventually reaching a value of zero. In contrast, the AQI for CO demonstrates relative stability with minimal reduction, while the AQI for NO₂ remains consistently at zero. Additionally, Figure 4.5 illustrates a gradual incremental rise in temperature accompanied by a corresponding gradual decrease in humidity.

The system demonstrates precise measurement capabilities in effectively detecting and quantifying the reduction in PM_{2.5} and PM₁₀ concentrations.

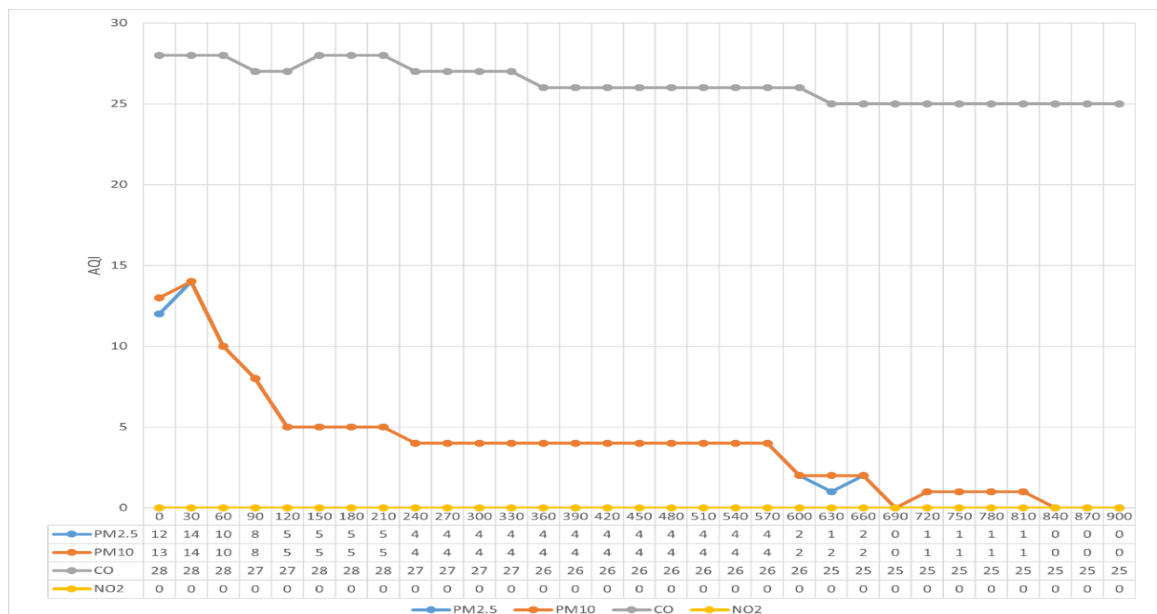


Figure 4.4: Air Quality Index (AQI) Trend Over Time in Sealed Enclosure

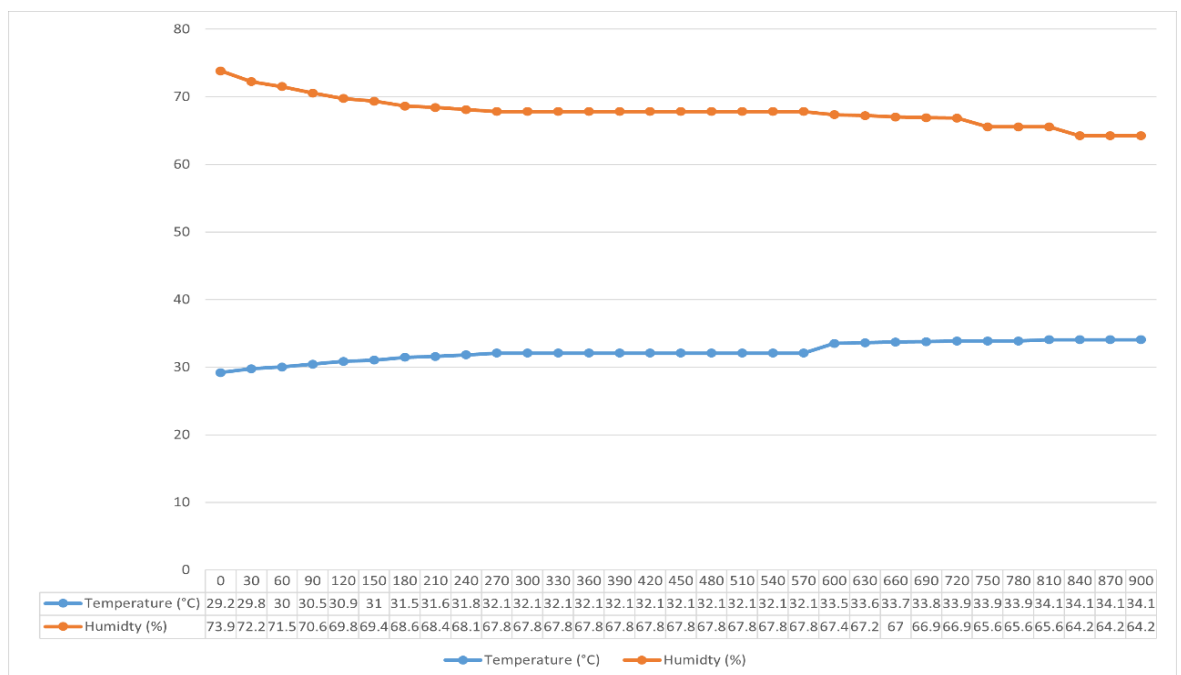


Figure 4.5: Temperature and Humidity Trend Over Time in Sealed Enclosure

4.3.2 Humidity Chamber Environment

The system is placed within a poorly ventilated space filled with steam to emulate a high-humidity environment. This simulation is maintained for a period of 15 minutes, during which measurements are captured at 30-second intervals.

Based on the data presented in Figure 4.6, it is observed that the Air Quality Index (AQI) of PM2.5 and PM10 experiences a sharp increase in the presence of steam, with PM2.5 showing a greater degree of increase compared to PM10. As the steam disperses, the AQI of PM2.5 and PM10 subsequently decreases. Conversely, the AQI of CO demonstrates a steady decrease, while the AQI of NO₂ consistently maintains a value of zero. In Figure 4.7, it is observed that the temperature remains relatively stable, while the humidity shows an increasing trend in the presence of steam. The humidity level rises significantly when the steam reaches its maximum level and then gradually decreases as the steam begins to disperse.

The system demonstrates exceptional measurement capabilities in accurately detecting and quantifying the fluctuations in humidity levels and the concentrations of particulate matter in the surrounding environment.

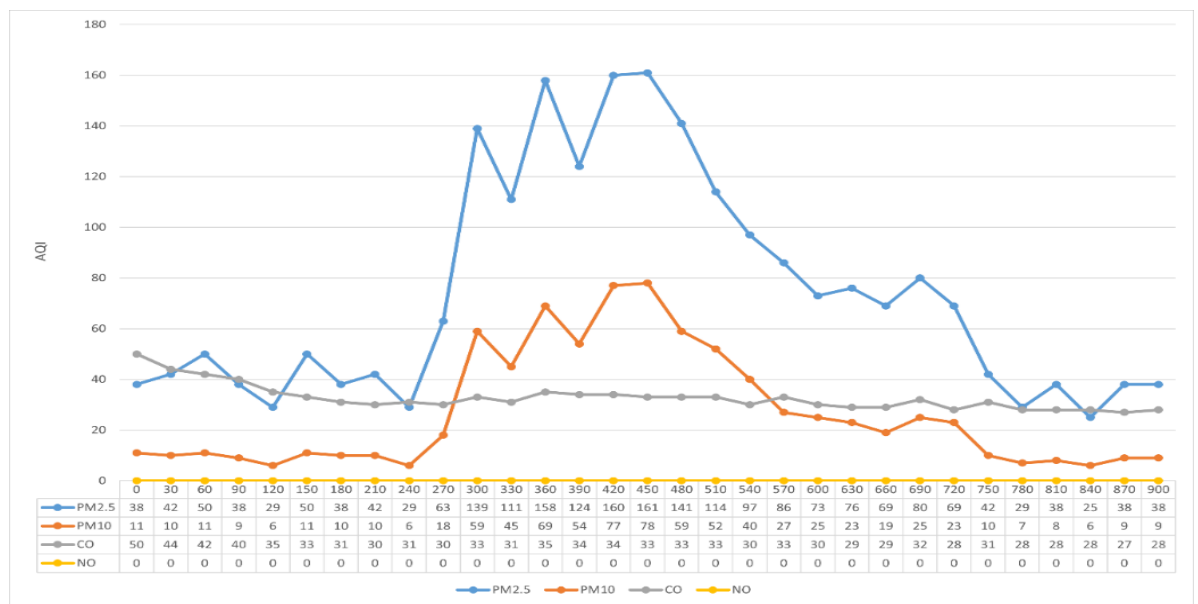


Figure 4.6: Air Quality Index (AQI) Trend Over Time in Humidity Chamber

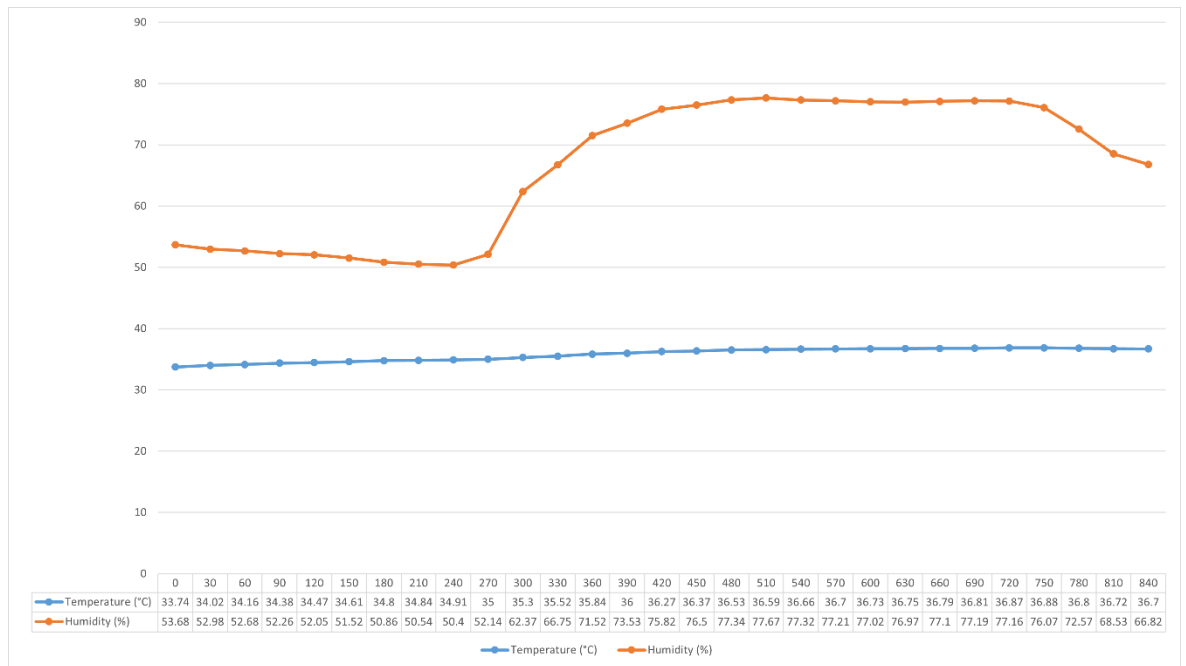


Figure 4.7: Temperature and Humidity Trend Over Time in Humidity Chamber

4.3.3 Smoke Box Environment

The system is placed within a confined and inadequately ventilated room, where a substantial sheet of burning paper is employed as the smoke source. This setup is maintained for a duration of 15 minutes, during which measurements are systematically recorded at 30-second intervals.

Based on the data presented in Figure 4.8, it is observed that the Air Quality Index (AQI) of PM_{2.5} and PM₁₀ undergoes an increase as the smoke spreads, eventually reaching hazardous levels. As the room becomes more ventilated, the AQI of PM_{2.5} and PM₁₀ starts to decrease. Notably, the AQI of PM₁₀ reaches higher levels compared to PM_{2.5} and exhibits a faster decrease during ventilation. On the other hand, the AQI of CO remains relatively stable with minimal fluctuations, while the AQI of NO₂ consistently maintains a value of zero. In Figure 4.9, the temperature shows a gradual increase, while humidity experiences a steady decline as the smoke spreads across the room.

The chart depicted in Figure 4.8 highlights the surpassing of the AQI threshold limit of 200, triggering the activation of the audible alarm (buzzer) until the AQI values drop below the established threshold. This demonstrates the system's capability to detect an escalation in the concentration of particulate matter (PM2.5 and PM10), specifically in this scenario representing smoke particles.

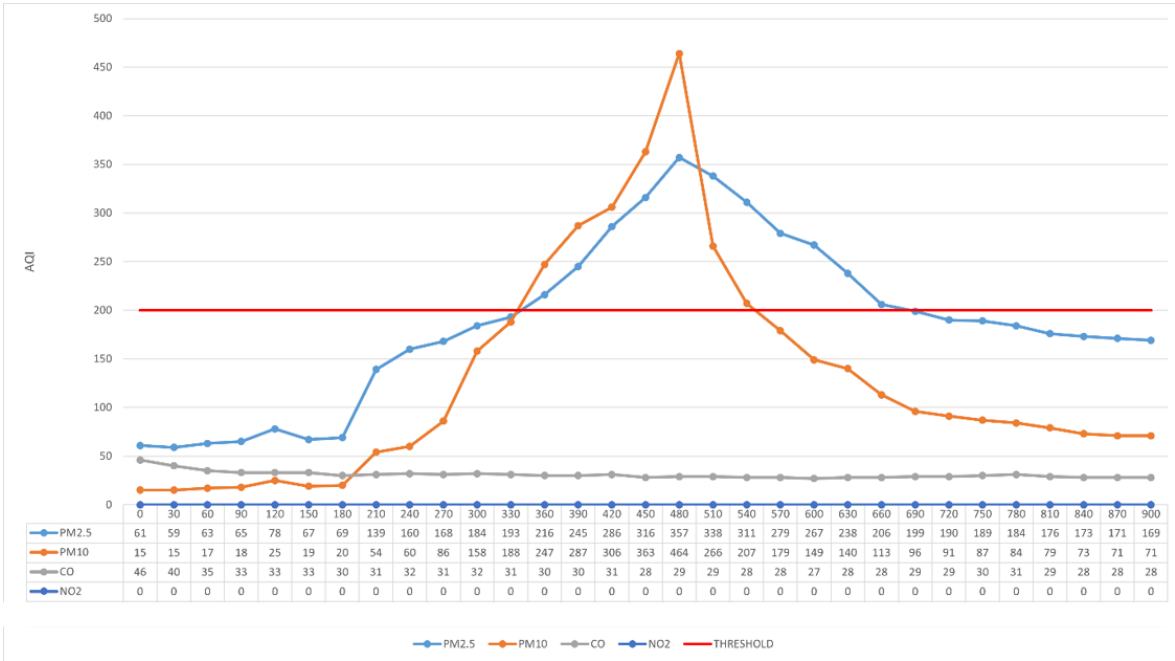


Figure 4.8: Air Quality Index (AQI) Trend Over Time in Smoke Box

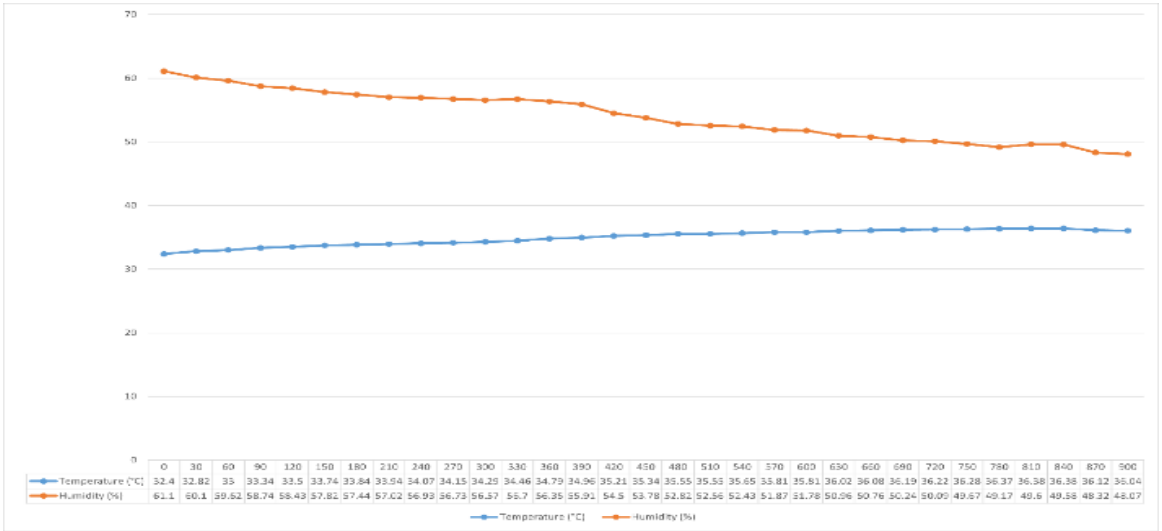


Figure 4.9: Temperature and Humidity Trend Over Time in Smoke Box

4.3.4 High Carbon Monoxide Environment

The system is exposed to an environment with an operational petrol-powered generator for a duration of 10 minutes, while measurements are captured at 30-second intervals.

Based on the data presented in Figure 4.10, it is observed that the Air Quality Index (AQI) of PM_{2.5} undergoes a sharp increase when the generator releases smoke but then gradually decreases towards its initial value. Similarly, the AQI of PM₁₀ experiences a milder increase compared to PM_{2.5} when the generator releases smoke and decreases towards its initial value. The AQI of CO shows an increasing trend, tending towards the threshold value. On the other hand, the AQI of NO₂ initially remains constant at a value of zero but immediately rises above the threshold value, activating the alarm system. In Figure 4.11, it is observed that the temperature exhibits a steady increase, while humidity steadily decreases.

The system demonstrates its ability to respond to variations in the concentration of PM_{2.5} and PM₁₀ when smoke is released by the generator, as well as to changes in the concentration of CO and NO₂ as the fumes from the generator continue to spread. This confirms that the system is capable of detecting variations in the concentration of carbon monoxide (CO) and nitrogen dioxide (NO₂).

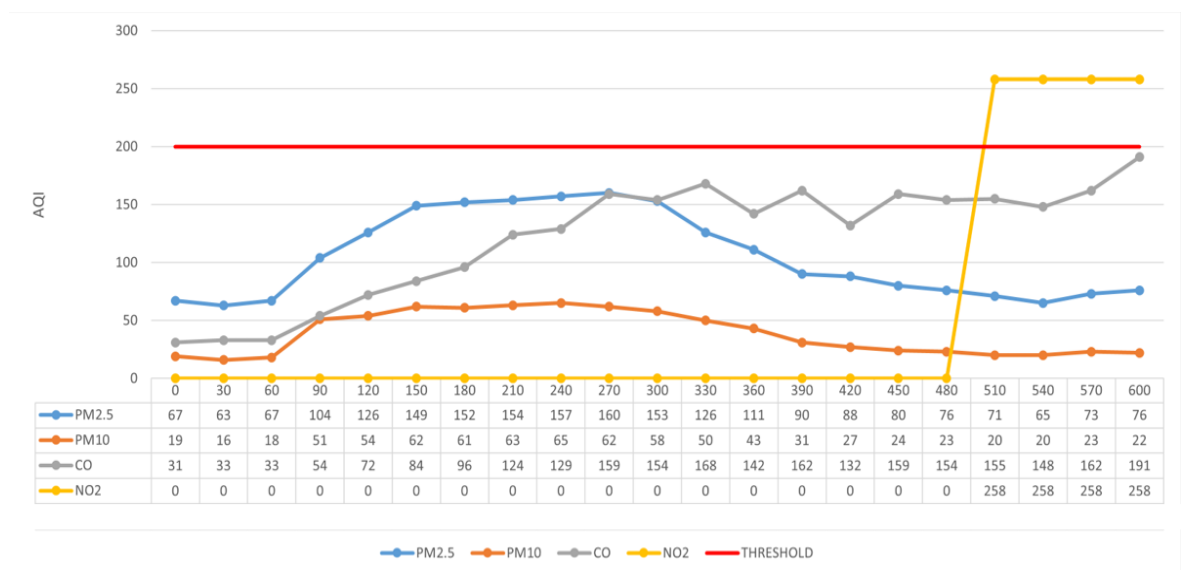


Figure 4.10: Air Quality Index (AQI) Trend Over Time for Carbon Monoxide (CO) Exposure

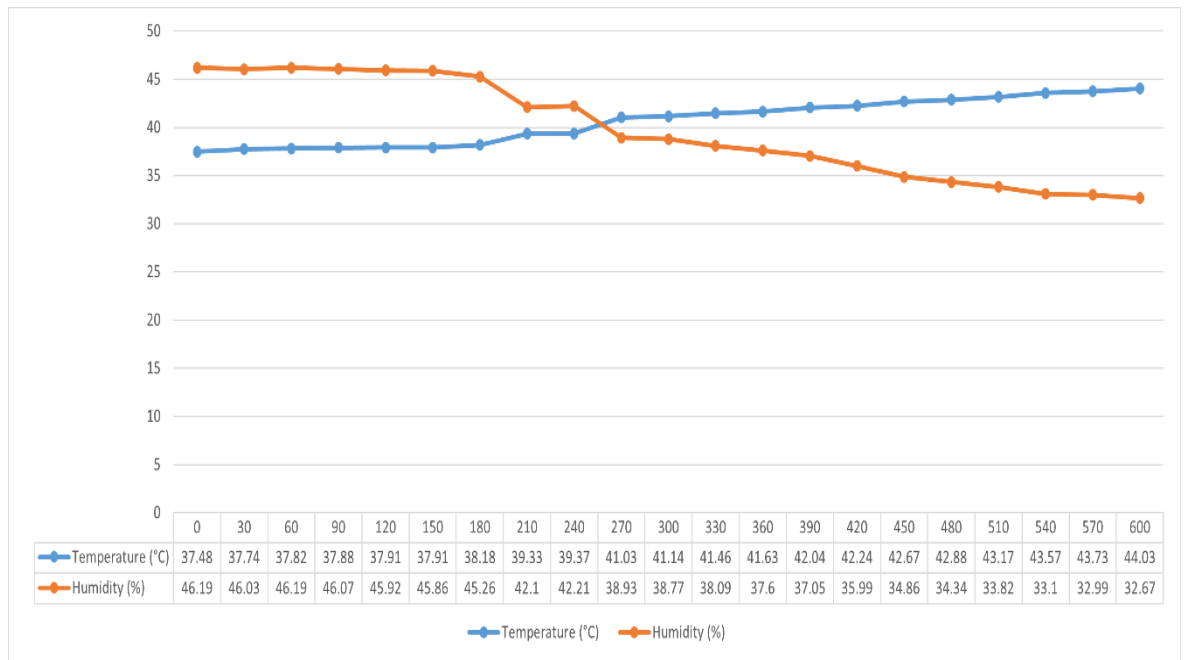


Figure 4.11: Temperature and Humidity Trend Over Time for Carbon Monoxide (CO) Exposure

4.4 Conclusion

The evaluation of the test reveals that the system demonstrates effective measurement capabilities in detecting changes in the concentration of all pollutants, including PM2.5, PM10, CO, and NO₂. Additionally, it proves proficient in capturing variations in environmental parameters such as temperature and humidity.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

The IoT-based air quality monitoring and pollution detection system project successfully achieved its objectives of developing a comprehensive and efficient solution for real-time air quality monitoring. Through the integration of various sensors, Arduino Nano as the ADC, ESP32 as the microcontroller, and the utilization of ThingSpeak platform for data visualization and analysis, the project successfully addressed the need for accurate and timely monitoring of air pollutants.

The system demonstrated its capability to measure the concentrations of pollutants such as PM2.5, PM10, CO, and NO, as well as environmental parameters including temperature and humidity. By providing real-time measurements, users were able to obtain immediate insights into the air quality conditions within their surroundings. The implementation of the alarm system was a crucial feature of the project, as it provided a timely warning when the Air Quality Index (AQI) surpassed the predefined threshold of 200. This ensured that users were promptly notified in case of deteriorating air quality, allowing them to take appropriate actions to protect their health and well-being. The integration of the LCD display enabled users to view the sensor readings directly on the device, providing a convenient and user-friendly interface for monitoring air quality data.

Throughout the development process, rigorous testing was conducted to ensure the system's accuracy, reliability, and functionality. The system underwent testing in varying air quality conditions, demonstrating its ability to track changes in air quality effectively. Testing also included simulating high-humidity environments and validating the system's performance under such conditions.

The project successfully delivered a fully operational IoT-based air quality monitoring and pollution detection system. The system's capabilities in real-time monitoring, alarm triggering, data visualization, and transmission to the ThingSpeak platform provided users with a comprehensive solution for monitoring and managing air quality.

In conclusion, the IoT-based air quality monitoring and pollution detection system project achieved its objectives by developing a reliable, user-friendly, and comprehensive solution for monitoring air quality. The successful implementation of the system, along with its demonstrated functionality and accuracy, lays the foundation for future advancements in the field of environmental monitoring and public health.

5.2 Recommendations

Based on the project's findings and experience gained during the implementation of the IoT-based air quality monitoring and alarm system, the following recommendations can be made for future implementations:

1. Integration of machine learning and artificial intelligence techniques to enhance data analysis and interpretation. By leveraging advanced algorithms, it becomes possible to identify patterns, predict air quality trends, and provide more accurate and insightful information to users.
2. Extending the implementation of the air quality monitoring system to cover multiple geographical locations. This would allow for comparative analysis of air quality across different regions and facilitate a broader understanding of pollution patterns and their underlying causes.
3. Development of a dedicated mobile application that connects to the IoT-based system, enabling users to access real-time air quality data, receive notifications, and track historical trends conveniently on their smartphones. The application can provide personalized recommendations for reducing exposure to poor air quality and promoting healthier lifestyles.
4. Explore integration possibilities with smart home systems to enhance automation and control features. This could include integrating the air quality monitoring system with smart thermostats, air purifiers, or ventilation systems, enabling automatic adjustments based on real-time air quality data.

REFERENCES

Air pollution. (2019, July 30). <https://www.who.int/health-topics/air-pollution>

Air pollution in Nigeria: Causes, Effects and Solutions. (2018, October 27). N99 And CE Air Pollution Masks | ATC MASK. <https://www.atcmask.com/blogs/blog/air-pollution-in-nigeria>.

Lewis, A. C., Lee, J. D., Edwards, P. M., Shaw, M. D., Evans, M. J., Moller, S. J., ... & White, A. (2016). Evaluating the performance of low cost chemical sensors for air pollution research. *Faraday discussions*, 189, 85-103.

Rai, A. C., Kumar, P., Pilla, F., Skouloudis, A. N., Di Sabatino, S., Ratti, C., ... & Rickerby, D. (2017). End-user perspective of low-cost sensors for outdoor air pollution monitoring. *Science of The Total Environment*, 607, 691-705.

Saad, S. M., Shakaff, A. Y. M., Saad, A. R. M., Yusof, A. M., Andrew, A. M., Zakaria, A., & Adom, A. H. (2017, March). Development of indoor environmental index: Air quality index and thermal comfort index. In *AIP Conference Proceedings* (Vol. 1808, No. 1, p. 020043). AIP Publishing LLC.

Saxena, P., & Sonwani, S. (2019). *Criteria air pollutants and their impact on environmental health* (Vol. 1, No. 1). Springer Singapore.

World Health Organization. (2021). *WHO global air quality guidelines: particulate matter (PM_{2.5} and PM₁₀), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide: executive summary*.

Particulate Matter (PM) Pollution | US EPA. (2023, January 6). US EPA. <https://www.epa.gov/pm-pollution>

Fine Particles (PM_{2.5}) Questions and Answers. (n.d.). <https://www.health.ny.gov/environmental/indoors/air>

Kampa, M., & Castanas, E. (2008). Human health effects of air pollution. *Environmental pollution*, 151(2), 362-367.

White, R. M., Paprotny, I., Doering, F., Cascio, W. E., Solomon, P. A., & Gundel, L. A. (2012). Atmospheric Monitoring.

Snyder, E. G., Watkins, T. H., Solomon, P. A., Thoma, E. D., Williams, R. W., Hagler, G. S., ... & Preuss, P. W. (2013). The changing paradigm of air pollution monitoring. *Environmental science & technology*, 47(20), 11369-11377.

Al Ahasan, M., Roy, S., Saim, A. H. M., Akter, R., & Hossain, M. (2018). Arduino-Based real time air quality and pollution monitoring system. *International Journal of Innovative Research in Computer Science & Technology (IJIRCST)* Volume-6, Issue-4.

Kaur, N., Mahajan, R., Bagai, D., & Student, P. G. (2016). Air quality monitoring system based on Arduino microcontroller. *International Journal of Innovative Research in Science, Engineering and Technology*, 5(6), 9635-9646.

Shah, H. N., Khan, Z., Merchant, A. A., Moghal, M., Shaikh, A., & Rane, P. (2018). IOT based air pollution monitoring system. *International Journal of Scientific & Engineering Research*, 9(2), 62-66.

Jha, R. K. (2020, July). Air quality sensing and reporting system using IoT. In 2020 Second international conference on inventive research in computing applications (ICIRCA) (pp. 790-793). IEEE.

Kumar, S., & Jasuja, A. (2017, May). Air quality monitoring system based on IoT using Raspberry Pi. In 2017 International conference on computing, communication and automation (ICCCA) (pp. 1341-1346). IEEE.

Kiruthika, R., & Umamakeswari, A. (2017, August). Low cost pollution control and air quality monitoring system using Raspberry Pi for Internet of Things. In 2017 International conference on energy, communication, data analytics and soft computing (ICECDS) (pp. 2319-2326). IEEE.

Marinov, M. B., Topalov, I., Gieva, E., & Nikolov, G. (2016, May). Air quality monitoring in urban environments. In 2016 39th International Spring Seminar on Electronics Technology (ISSE) (pp. 443-448). IEEE.

Gunawan, T. S., Munir, Y. M. S., Kartiwi, M., & Mansor, H. (2018). Design and Implementation of Portable Outdoor Air Quality Measurement System using Arduino. *International Journal of Electrical & Computer Engineering* (2088-8708), 8(1).

Abiduzzaman, S. M., Mansor, H., Gunawan, T. S., & Ahmad, R. (2021, August). Real-Time Outdoor Air Quality Monitoring System. In *2021 IEEE 7th International Conference on Smart Instrumentation, Measurement and Applications (ICSIMA)* (pp. 140-145). IEEE.

Phala, K. S. E., Kumar, A., & Hancke, G. P. (2016). Air quality monitoring system based on ISO/IEC/IEEE 21451 standards. *IEEE Sensors Journal*, 16(12), 5037-5045.

Simu, S., Turkar, V., Martires, R., Asolkar, V., Monteiro, S., Fernandes, V., & Salgaoncary, V. (2020, December). Air Pollution Prediction Using Machine Learning. In *2020 IEEE Bombay Section Signature Conference (IBSSC)* (pp. 231-236). IEEE.

APPENDIX

APPENDIX A: AQI CALCULATOR

```
#include <Arduino.h>

#include "AQI_Calc.h"

/**
 * @brief Constructor for the AQI_Calc class.
 */

AQI_Calc::AQI_Calc(void)

//Assign the range and breakpoint for each pollutant
: PM10 {{0,54,0,50},{55,154,51,100},
        {155,254,101,150},{255,354,151,200},
        {355,424,201,300},{425,504,301,400},
        {505,604,401,500}}},
  PM2_5 {{0.0,15.4,0,50},{15.5,40.4,51,100},
        {40.5,65.4,101,150},{65.5,150.4,151,200},
        {150.5,250.4,201,300},{250.5,350.4,301,400},
        {350.5,500.4,401,500}}},
  CO   {{0.0,4.4,0,50},{4.5,9.4,51,100},
        {9.5,12.4,101,150},{12.5,15.4,151,200},
        {15.5,30.4,201,300},{30.5,40.4,301,400},
        {40.5,50.4,401,500}}},
  NO2  {{0.0,0.0,0,50},{0.0,0.0,51,100},
        {0.0,0.0,101,150},{0.0,0.0,151,200},
        {0.65,1.24,201,300},{1.25,1.64,301,400},
        {1.65,2.04,401,500}}
```

```

{
    //Initialize private variables

    Pollutant[P_PM10] = PM10;

    Pollutant[P_PM2_5] = PM2_5;

    Pollutant[P_CO] = CO;

    Pollutant[P_NO2] = NO2;
}

/**
 * @brief Method to get the AQI level of a pollutant
 * @param pollutant Pollutant for which the AQI level is to be calculated
 * @param pollutantConc Concentration of the pollutant
 * @return AQI level of the pollutant
 */
AQI_LEVEL AQI_Calc::GetAQIlevel(POLLUTANTS pollutant,float pollutantConc)
{
    AQI_LEVEL level = INVALID;

    switch(pollutant)
    {
        case P_PM10:
            for(uint8_t i = 0; i < NO_OF_AQI_LEVELS; i++)
            {
                if(pollutantConc >= PM10[i].BP_Low && pollutantConc <= PM10[i].BP_High)
                {
                    level = (AQI_LEVEL)i;
                }
            }
        }
    }

```

```

    }
break;

case P_PM2_5:

    for(uint8_t i = 0; i < NO_OF_AQI_LEVELS; i++)
    {
        if(pollutantConc >= PM2_5[i].BP_Low && pollutantConc <= PM2_5[i].BP_High)
        {
            level = (AQI_LEVEL)i;
        }
    }
break;

case P_CO:

    for(uint8_t i = 0; i < NO_OF_AQI_LEVELS; i++)
    {
        if(pollutantConc >= CO[i].BP_Low && pollutantConc <= CO[i].BP_High)
        {
            level = (AQI_LEVEL)i;
        }
    }
break;

case P_NO2:

    for(uint8_t i = 0; i < NO_OF_AQI_LEVELS; i++)
    {
        if(pollutantConc >= NO2[i].BP_Low && pollutantConc <= NO2[i].BP_High)
        {
            level = (AQI_LEVEL)i;

```

```

    }

    }

    break;

}

return level;

}

/**
 * @brief Method to get the AQI Index of a pollutant
 * @param pollutant Pollutant for which the AQI level is to be calculated
 * @param pollutantConc Concentration of the pollutant
 * @return AQI Index of the pollutant
 */

float AQI_Calc::ComputeIndex(POLLUTANTS pollutant, float pollutantConc)
{
    AQI_LEVEL level;

    level = AQI_Calc::GetAQIlevel(pollutant, pollutantConc);

    switch(pollutant)
    {
        case P_PM10:

            PM10[level].Index = (pollutantConc - PM10[level].BP_Low)*

                ((PM10[level].I_High - PM10[level].I_Low)/

                (PM10[level].BP_High - PM10[level].BP_Low)) +

                PM10[level].I_Low;

            break;

        case P_PM2_5:

```

```

    PM2_5[level].Index = (pollutantConc - PM2_5[level].BP_Low)*
        ((PM2_5[level].I_High - PM2_5[level].I_Low)/
        (PM2_5[level].BP_High - PM2_5[level].BP_Low)) +
        PM2_5[level].I_Low;

break;

case P_CO:

    CO[level].Index = (pollutantConc - CO[level].BP_Low)*
        ((CO[level].I_High - CO[level].I_Low)/
        (CO[level].BP_High - CO[level].BP_Low)) +
        CO[level].I_Low;

break;

case P_NO2:

    NO2[level].Index = (pollutantConc - NO2[level].BP_Low)*
        ((NO2[level].I_High - NO2[level].I_Low)/
        (NO2[level].BP_High - NO2[level].BP_Low)) +
        NO2[level].I_Low;

break;

}

return Pollutant[pollutant][level].Index;

}

```

APPENDIX B: BME 280

```
#include <Arduino.h>

#include "bme280.h"

/**
 * @brief Constructor.
 *
 * Initializes the BME280 sensor using the provided I2C address.
 *
 * @param addr: I2C address of the BME280 sensor.
 */
BME::BME(uint8_t addr)
{
    this->deviceAddr = addr;
    bme.begin(this->deviceAddr);
}

/**
 * @brief Read temperature or humidity data from the BME280 sensor.
 *
 * @param value: The type of data to read (temperature or humidity).
 *
 * @return The temperature or humidity data, as a floating point value.
 */
float BME::GetValue(uint8_t value)
{
    switch(value)
    {
        case BME::TEMP:
```

```

        return bme.readTemperature();

    case BME::HUM:
        return bme.readHumidity();
    }
}

```

APPENDIX C: MiCS-6814

```

#include <Arduino.h>

#include "mics6814.h"

/**
 * @brief Constructor.
 * @param NO2Pin: Analog pin number for reading NO2 data.
 * @param NH3Pin: Analog pin number for reading NH3 data.
 * @param COPin: Analog pin number for reading CO data.
 */
MICS6814::MICS6814(uint8_t NO2Pin,
                  uint8_t NH3Pin,
                  uint8_t COPin)
{
    this->NO2Pin = NO2Pin;
    this->NH3Pin = NH3Pin;
    this->COPin = COPin;
}

```

```

pinMode(this->NO2Pin, INPUT_PULLUP);
pinMode(this->NH3Pin, INPUT_PULLUP);
pinMode(this->COPin, INPUT_PULLUP);
}

/**
 * @brief Read gas data from the MICS6814 sensor.
 * @param value: The type of gas data to read (NO2, NH3, or CO).
 * @return The gas data, as a 16-bit unsigned integer in parts per million (ppm).
 */
uint16_t MICS6814::GetValue(uint8_t value)
{
    uint16_t gasData;
    switch(value)
    {
        case GAS::NO2:
            gasData = analogRead(this->NO2Pin);
            return map(gasData,0,1023,0.00,2.04);
        case GAS::NH3:
            gasData = analogRead(this->NH3Pin);
            return map(gasData,0,1023,1,500);
        case GAS::CO:
            gasData = analogRead(this->COPin);
            return map(gasData,0,1023,1,1000);
        default:
            return 0;
    }
}

```

```
}  
}
```

APPENDIX D: MQ7

```
#include <Arduino.h>
```

```
#include "MQ7.h"
```

```
MQ7::MQ7(uint8_t analogPin)
```

```
{  
  this->analogPin = analogPin;  
  this->R0 = 40000.0;  
}
```

```
float MQ7::GetPPM(void)
```

```
{  
  uint16_t sensorRawValue = analogRead(this->analogPin);  
  float sensorVolt = sensorRawValue/1024.0*5;  
  float RS_gas = (5.0-sensorVolt)/sensorVolt;  
  float ratio = RS_gas/this->R0;  
  float x = 1538.46*ratio;  
  float ppm = pow(x,-1.709);  
  return ppm;  
}
```

